DRS BOSTON 2024 JUNE 24 — 28

Rethinking design: Prototyping sustainable futures in everyday life

Mia Hesselgren*, Sara Ilstedt, Martin Sjöman

KTH School of Industrial Engineering and Management, KTH Royal Institute of Technology, Stockholm, Sweden

*Corresponding author e-mail: miahes@kth.se

https://doi.org/10.21606/drs.2024.273

Abstract: To support sustainability transitions several different design approaches are required. In a series of Designerly Living Labs, we have prototyped possible sustainable futures in the context of people's everyday lives to explore the complexities of lifestyle changes and socio-technical system shifts. Together with users engaged as reflexive co-researchers, we have explored potential new practices and uncovered system-level tensions and deficiencies. System changes have been initiated by engaging relevant actors in learning processes and by bridging learning to decision-makers. However, the approach requires some rethinking of design as a practice. Instead of supporting people's current needs and lifestyles, we design for sustainable futures that users do not yet request. This emerging design practice challenges traditional co-design and user participation methods, leading to new ethical considerations.

Keywords: Future prototyping, Everyday life, Designerly Living Labs, Sustainability transitions

Introduction

The work described in this paper is part of a growing design research area seeking to understand how design can support sustainability transitions. Sengers et al. define a sustainability transition experiment as "an inclusive, practice-based and challenge-led initiative, which is designed to promote system innovation through social learning under conditions of uncertainty and ambiguity" (Sengers et al., 2019, p. 161). Such research is complex and involves a high degree of uncertainty since sustainability innovations need to be considered socio-technical, i.e. they require changes in social practices and cultural meanings, product and service systems, industries and markets, and policy and regulations (Elzen et al., 2004). Many have argued that open-ended, real-world experimentation is required to understand the potential outcomes of these changes (Caniglia et al., 2021; Ceschin, 2014; Edwards & Bulkeley, 2017; Loorbach, 2007) and that such experimentation should take place in the context of everyday

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International Licence.

lives (Dourish & Button, 1998; Irwin et al., 2015; Liedtke et al., 2015; Spaargaren, 2011). Open-ended experiments have also been found to engage users in 'second-order learning', allowing them to question their current practices (Geels et al., 2019). Consequently, in many research fields, there is an ongoing experimental turn (Corsin Jiménez, 2014; Kimbell & Bailey, 2017; McCrory et al., 2020) with several different approaches and methodologies, including, most notably, living labs.

In this paper, we describe three design-driven or 'designerly' living labs. These living labs were extensive full-scale experiments targeting three different consumption categories: everyday mobility, workplace commuting, and housing. In all three cases, design practice was used to prototype or stage possible sustainable futures in the context of everyday life, engaging user participants in explorative learning processes. Notably, the prototypes were not designed to support present lifestyles but explicitly designed to challenge them.

Several design approaches can be categorised as designing for sustainability transitions. We suggest that for designers to contribute to sustainability innovations, there is a need for approaches that build on open-ended experimentation in everyday-life contexts and that challenge people to replace unsustainable behaviours and lifestyles. Inspired by social practice theory and backcasting, the designerly living labs presented in this paper exemplify such an approach, where designers do not uncritically design what people want without sustainability considerations. However, we have found that this type of research has come with specific challenges and gains. To challenge users to experiment and re-learn, we have found it fruitful to engage them as active and reflective *co-researchers*. In doing so, our research has provided actionable results, identifying barriers and pathways towards envisioned sustainable futures.

Background and related research approaches

Ceschin and Gaziulusoy (2016) distinguish between design for sustainability approaches with insular or systemic aims. Examples of more insular approaches are Eco-Design (Tukker et al., 2000), which aims to reduce the adverse environmental effects of products, and Design for Sustainable Behavior (Lilley, 2009; Lockton et al., 2008;), which mainly aims to change unsustainable behaviours by nudging, informing or guiding users. At the other end of the scale are approaches that address more systemic issues. Burns et al. (2006) proposed the term transformation design for using design-led innovation to tackle social and economic issues. Irwin et al. (2015) suggested that the term transition design refers to design that aims to change complex systems, which we refer to in this paper.

Overall, it has been suggested that transition design should aim to understand the local contexts where change is needed and involve the people affected by the proposed change (Irwin et al., 2015). Consequently, participatory design and co-design have become important. Research approaches based on participatory design and social innovation have, to a greater extent, been carried out within real-world contexts, mainly focusing on social innovation and

emphasising design as a way to 'activate' citizens and amplify grassroots solutions (see, e.g. Hillgren et al., 2011; Manzini, 2015; Manzini & Rizzo, 2011).

In parallel, the last fifteen years have seen the rise of design fiction (Sterling, 2009), critical design (Dunne & Raby, 2011), speculative design (Auger, 2013), and design visioning (Gaziulusoy & Ryan, 2017). These approaches often utilise future scenarios or speculative visualisations to facilitate co-design workshops and user or stakeholder co-creation. Early examples include co-creating Design-Orienting Scenarios to reduce household consumption in the SUSHouse project (Green & Vergragt, 2002). Later examples include highly developed design tools to facilitate policy co-creation (Gaziulusoy & Ryan, 2017; Lähteenoja et al., 2023). Co-design workshops allow for exploring radical concepts but do not necessarily confront the complexity and entanglements of everyday life, social interactions and societal structures.

Living labs have emerged as research structures for various real-life experimentation approaches. Even though living labs often emphasise user co-creation (ENOLL, n.d.), there is great variety and many unclarities regarding how users are engaged (Hossain et al., 2019; Puerari et al., 2018; Shuurman et al., 2015). While more technologically oriented living labs tend to focus mainly on sourcing user data for testing and evaluating ICT solutions (Kommonen & Botero, 2013; Leminen et al., 2014; Marres & Stark, 2020), other living labs are more oriented towards sustainability and societal change, using more qualitative approaches. Urban Living Labs (ULL) (Marvin et al., 2018; Voytenko, 2016) often seek to foster stakeholder or community engagement to address local problems. Similarly, in participatory design and social innovation, several living labs have included experimentation in real-life settings, using design to activate or enable community engagement (see, e.g. Fassi et al., 2013; Hillgren, 2013; Manzini, 2015). Other ULLs use ethnographic methods to test socio-technical configurations in real-world settings. In these cases, Brons et al. (2022) suggest a distinction between living labs that utilise broad but mostly un-reflexive user engagement to *understand* daily life and living labs that engage users more deeply in *breaking with* daily life. The latter type is, however, exemplified by co-designing and visualising hypothetical future scenarios in workshop settings. In summary, we find few living labs that explore breaking with daily life employing more reflexive user engagement in the context of everyday life settings.

To include social and societal structures, practice-oriented design (Kuijer, 2014; Shove & Watson, 2006) suggests that designers broaden their focus from insular products and individual users to instead engage with social practices and how these are shaped by norms and images, material structures, and individual skills (Reckwitz, 2002). By analysing the elements of a practice, designers identify 'points for intervention' that may be on higher system levels, like norms or regulations. Some living lab experiments have built on such practice-oriented design interventions, notably the experiments on ways to reduce domestic water consumption by designing and diffusing new 'proto-practices' to replace the practice of daily hot showers (Kuijer, 2014; Scott et al., 2012).

As discussed by Ceschin and Gaziulusoy (2019), without designing changes in the real world, designing for sustainability transitions may be limited to analysis and "the communication of high-level outcomes through visualisations" (p 132). Similarly, to provide action-oriented knowledge for sustainability, Caniglia et al. (2021) highlight *contextual realisation* for experimenting with new social configurations and alternative social norms. Overall, we see a need for complementary design research approaches that challenge existing norms and practices hands-on and do this in everyday life. Since 2012, our research group has conducted several such sustainability experiments to explore potential barriers and pathways for sustainability transitions. Here, design interventions have enabled engaged and reflexive user participants to learn from experimenting with new, more sustainable practices. We call these experiments Designerly Living Labs (DLL).

Designerly living labs - Our research approach

Reductions in consumption require finding alternative yet acceptable ways of living. In the Designerly Living Labs (DLL), we have designed products, services, policies and facilities that could be part of sustainable futures. These future concepts have been prototyped and explored in the contexts of people's everyday lives and tried out over relatively long periods. As this research approach has been used chiefly in early research stages, the DLL concepts have been deliberately *provisional* and designed as tools for learning instead of employing pre-studies or engineering processes to determine optimal solutions (Sjöman & Hesselgren, 2020).

Our approach has been inspired by backcasting as a method to visualise a future goal and then develop a roadmap to reach this goal (Ilstedt & Wangel, 2014). This way, the DLL future prototyping has been viewed as a first step in an iterative development process. The DLLs were used in the early phases of concept development, and the experimentation provided insights that were taken forward to relevant stakeholders. Through the demonstrations of sustainable lifestyles and actively staging and interacting with the future prototypes, several stakeholders have been inspired to initiate subsequent developments and projects independently.

We do not engage citizens to co-design future concepts or seek to activate them in grassroots initiatives or bottom-up community actions. Instead, we ask them to reflexively explore the future concepts and experiment with integrating them in their everyday lives (Sjöman, 2023). Hence, we have engaged the participants as co-researchers rather than co-designers. To let new practices emerge, time for experimenting, re-learning and shaping new routines has been required. Therefore, an important aspect of the labs is that they have all been longitudinal, lasting from several months to more than a year. To study and understand evolving practices, barriers, and motivations for change, we have continuously followed up with interviews, observations, and quantitative data collection during the experiments. In several cases, trigger materials have been designed to support reflectivity in interviews, and research devices such as diaries have been used to catch pain points or

breakdown moments and to learn from them (Hesselgren, 2023). This has produced insights into the co-researchers' experiences: the challenges and problems they face, persistent habits and perceptions, positive experiences and benefits, new practices and unexpected consequences.

Method

The empirical results of the labs and the data collected are vast and have been analysed from different perspectives in several iterations to identify similarities and differences. Two of the authors have analysed the DLLs, other living labs and real-life experiments as part of their doctoral dissertations. This was carried out through developing a design program (see Hesselgren, 2019) and cross-case analysis (see Sjöman, 2023). Previous publications are referenced in the following three case descriptions for more details about the data collection and analysis in each specific DLL.

The three aspects we present in this paper have been selected since they are visible in all the labs and were significant for the approach and results. These aspects were also deemed to be of interest for design research.

Three example cases - three real-life experiments

In this section, we present the three research cases. We briefly describe their background and purpose, the design of the experiments, the participants' explorations and learnings, and the project results and outcomes.

'A Car-Free Year' (henceforth 'Car-Free')

In the research project 'A Car-Free Year', three families with children in Stockholm replaced their car trips by using light electric vehicles: electric bikes, box bikes, scooters, and four-wheeled motorcycles, as well as public transport and active mobility. The families' experiences of car-free living were followed throughout the year. Their experiences formed the basis for concepts on how a city such as Stockholm could welcome more car-free families with children (for further details, see Hesselgren & Hasselqvist, 2016; Hasselqvist & Hesselgren, 2019).

For the participants to engage in car-free living, each family selected light-electric vehicles made available for them to rent (see example in Figure 1). Furthermore, the families were supported by these material elements and empowered to gain new knowledge, such as how to dress when cycling in the winter. Included in the project were, for example, allowed to buy winter gear, but they also had access to free bike repairs and tyre changes, which proved essential to sustaining the winter cycling practice. In this process, the families were engaged in co-designing what worked for them to carry out the intervention in their everyday situations.



Figure 1 One of the participating families with their electric vehicles.

The families' travel practices and bundled practices, for example, doing the weekly grocery shopping, were explored through monthly interviews. Furthermore, the adults kept a travel diary describing their experiences, emotions, and reflections. Each family was also allowed twelve car trips during the year, registered on a scorecard. This simple learning device provided important insights into situations when car-free living was impossible. The travel diary was designed as a trigger material to spark discussions in the interviews and proved to help the participants reflect deeper on their experiences and emotions. With time, it became more comfortable for the participants to share their thoughts on more complex issues, such as parenthood, where they, for example, felt that they were challenging the norm of being a good parent in their communities by not having a car.

During the analysis phase of the project, the experiences that the participants had expressed were interpreted, and we, as a research team, learned from these. Following, concepts for how car-free living could be developed were designed. These concepts were created collaboratively, engaging the families in workshops to discuss the suggestions. Collaborative workshops were also conducted with stakeholders from civil society and the public and private sectors.

The developed concepts were visualised as 2D images that balanced concrete aspects of the families' experiences with more abstract ideas on future possibilities. This balancing was intended to stimulate and trigger ideas of future possibilities when brought to decision-makers. As a complement, videos were also produced where the families' stories were told by themselves and where the research team shared insights on car-free living. These designed materials were used to bridge the learnings across to decision-makers, enabling new learning and possible starting points for actions. The visualisations and films were shown and discussed in local, regional and national meetings to discuss sustainable mobility solutions and how to welcome more families to live car-free.

'Work Closer'

In 'Work Closer', a 'work hub' or a 'Neighborhood Telecommuting Centre' was built in a villa suburb 35 km outside Stockholm, Sweden. The living lab was initiated before the Covid-19

pandemic and was mainly intended to reduce the carbon footprint of residents with long commutes. The work hub was not designed based on research about local user needs but was quickly staged as an experiment to explore the larger socio-technical picture around it. There was, for example, an interest in changes in transport modes, health aspects, and possible rebound from driving more in leisure time.

An office space was rented at a central location, remodelled and fitted with 16 desks and computer screens, a well-equipped meeting room and three 'Zoom booths' for remote meetings (see Figure 2). Two large employers that were project partners agreed to let employees living in the area use the service prototype but found that many practical issues had to be solved. This work took several months, including resolving security issues, insurance policies, teamwork and management routines.



Figure 2 One of the areas at the work hub.

Many outcomes were unexpected. This was before the Covid-19 outbreak, and it became clear that very few people were allowed to work remotely more than one day per week. Even those who were allowed to and had a clear ambition to use the work hub found that things were always stopping them, such as a new project, an important meeting, or supporting a colleague. 67 participants signed up for the work-hub experiment, but only around 25 came to use it more than a few times, and less than 10 used it regularly for a more extended period. Instead of exploring ways to reduce commuting, we ended up identifying many barriers to remote work. Even when working with experienced architects, several issues were not predicted. The well-equipped meeting room was never used, since unlike the coworking spaces in the city centre, the closer colleagues remained at the main office. Instead, many participants spent several hours daily meeting their colleagues remotely, and the three Zoom booths were insufficient.

The participants who could work remotely experienced reduced stress and more time with their families. The few who used the work hub several days per week also gradually made other changes, such as shopping locally and purchasing e-bikes. One participant felt such a substantial relief working locally that she quit her job and found a new line of work closer to home (for further details, see Vaddadi et al., 2022).

The project received much attention. It was featured on TV and visited by a secretary of state before it came to an unexpected halt with the outbreak of Covid-19. In a few weeks, all the identified barriers to remote work were dealt with or simply ignored when almost all participants were required to work only from home. However, the experience of interacting with the prototype led the two large employers to initialise changes to their operations. After the pandemic, the experiment was extended for a year, run by a start-up company with funding from the municipality.

'CoKitchen'

'CoKitchen' aimed to develop sustainable shared student housing for the future. For the last 25 years, student housing in Sweden has been almost entirely small one-room flats that are expensive, resource-intensive, and do not support social connections. Academic Housing, the governmental student housing company, had been commissioned to build 5800 new student housing and wished to explore more sustainable alternatives.

The project was completed in a building a permit-free testbed at the university campus. During the project, the testbed was rebuilt twice with different floor plans, and after each rebuild, 4-5 students lived there for one academic year. Contextual interviews and observations were made 2-3 times per year. When the project started, the lab was built as four one-room flats of 21 m2. In the first rebuild, we knocked out the walls between the studios to form a large combined kitchen and living room, surrounded by four bedrooms with private baths. The year after, a completely new co-living apartment was designed, with small private bedrooms (8m2), two shared bathrooms, a large kitchen and a small separate living room, in total 70 m2 shared area (see Figures 3 and 4). After both rebuilds, we made different design interventions to promote sustainable practices, prevent conflicts, mainly about cleaning, and help build community in the group. We asked the students to share a vegetarian dinner every other week and designed a shared cleaning schedule and recycling bags.



Figure 3 The three different floor plans of the testbed flat.



Figure 4 The large kitchen in the second rebuild.

A common theme in both experiments was that the students said they became a family. They celebrated each other's birthdays and helped out when one student got ill with Covid-19. Always having somebody around and having spontaneous social encounters was very appreciated. We found that sustainable practices spread among the students, such as eating more plant-based food, recycling, hang-drying, and recycling (for further details, see Ilstedt & Sjöman, 2022). The students, as co-researchers, were encouraged to improve their home and the cleaning schedule to fit their routines.

Climate calculations were made comparing typical one-room student flats with shared apartments. Building shared housing uses 25-35% less CO2 as many facilities are shared, but also since more people can live in the building. The building costs were 20-25% lower for the same reasons (Malmqvist & Brismark, 2023). Another finding was the many norms, policies and rules that obstruct or make shared housing unnecessarily expensive, such as building and accessibility regulations.
Spreading and bridging the results to stakeholders was integral to the project. CoKitchen was featured on television and radio, in daily papers, specialist journals, and in an innovation podcast. The final report was written and designed in a non-academic format to inspire reading and presented at an open seminar. Most importantly, Academic Housing redesigned a housing project from one-room flats to a diverse range of one-, two-, four- and six-bedroom apartments- By doing so, the number of students who could live in the building was increased from 250 to 390.

Findings from the cases

Identifying unexpected benefits through long-term experimentation

Rather than identifying and fulfilling users' perceived needs, we have aimed to design

concepts that challenge norms and behaviours. Given the participants' challenges, we found that an extended test period was essential to see how new behaviours emerge. Initially, significant re-learning is required; many people find the change unusual and uncomfortable, and it is easy to slip back into old behaviours (Ilstedt & Sjöman, 2022). After a while, changes may settle in the form of new, more sustainable practices that gradually become normalised. At this stage, participants have often discovered unexpected benefits that sometimes outweigh the perceived benefits of old practices. In Car-Free, the children learned to travel with public transportation as the parents could not drive them around. Over time, the children came to appreciate their new-won independence, and the parents gained time. In Co-Kitchen, the students greatly appreciated the big kitchen and living room and did not wish to replace them with a larger bedroom or a private bathroom. In Work Closer, there were indications that the few co-researchers who could use the work hub more often began to develop more local lifestyles and experienced increased well-being.

Overall, we find the cumbersome relearning and the immediate loss of convenience easy to foresee, but the long-term, often social gains are not. Previously, we have referred to this as passing the 'valley of inconvenience' (see Figure 5).

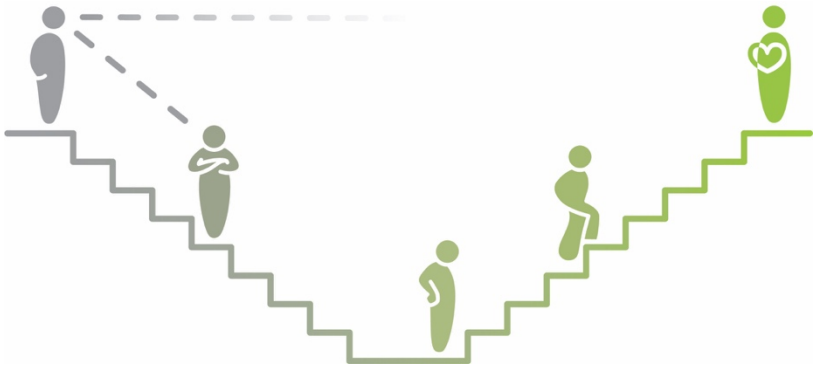


Figure 5 The Valley of Inconvenience (Sjöman, 2023). From our current viewpoint, it is easy to foresee the initial loss of convenience associated with re-learning. However, seeing what we may gain when new sustainable practices have settled and become normalised is hard.

Engaging co-researchers rather than co-designers.

The sustainable future prototypes in the labs were not immediately attractive solutions that anyone would choose or design themselves. The sustainable concepts often did not fit current norms and the participants' practices. Integrating the design interventions into their lives was often challenging and required significant relearning. We have therefore sought to engage users as actively engaged *co-researchers*. This has meant finding motivated participants, being transparent regarding the purpose of the research, and engaging them in a reflexive capacity. This, in turn, has required close researcher and co-researcher relations. The co-researchers were encouraged to make changes, learn and hack the experiments in order for them to work in their everyday life. As described in the case studies, the families in Car-Free had to learn how to cycle in the winter, students in CoKitchen developed extra storage

for the (intentionally) too-small kitchen and redesigned their cleaning schedule, and participants in Work-Closer had to find ways to cope with their colleagues' unwillingness to meet online.

Due to the long-term exploration, we have found it important to use *balancing strategies* to make the interventions 'work' for the participants while simultaneously provoking change and experimentation. In earlier publications, we have discussed balancing strategies such as combining more 'strict' interventions with highly motivated participants, as in Car-Free (Hesselgren et al., 2017), while in Work Closer, actively encouraging the participants to experiment through 'softer' strategies.

Revealing unfit structures on a socio-technical systems level

There are many advantages of real-life experimentation, but the projects are also very resource-demanding since they not only involve designing and researching, but also 'constructing' and setting up physical spaces, products, and services. Exploring and practically staging a sustainable future has often enabled the identification of system-level errors and glitches, conflicting rules, and regulations. In Work Closer, the process for the employer companies to allow remote work, though cumbersome in the project, provided important learnings both for these research partners and the research team. In CoKitchen, it became increasingly clear how regulations and established processes counteracted the building of shared housing. In all cases, we have found that system-level changes were required to allow the sustainability concepts to be more broadly implemented in society. These changes were regularly outside the scope of the design interventions but were still important findings. This way, the prototypes functioned as *learning devices* (Sjöman & Hesselgren, 2020), both in the staging of the prototype and during the use phase.

Discussion: Rethinking designer roles

Overall, the case learnings illustrate a core challenge of designing for sustainability transitions. On the one hand, a sustainable future society must provide for people's needs and lifestyles and support daily routines. On the other hand, we, as designers, cannot uncritically provide what people request and desire since this often boils down to more convenience, more consumption, and unchanged behaviours. Affluent middle-class consumers may request electric or biofuel cars but are generally unwilling to make more substantial lifestyle changes. It is easy to trust future technological developments to solve everything, but these will not be sufficient given the current developments. It is difficult for people to envisage alternatives to their routines or lifestyles; consequently, as Spaargaren (2011) notes, user-centric or user-driven design approaches such as HCD or co-design are likely inadequate.

Practice-Oriented Designing

Many have suggested that understanding social practices is critical when designing for sustainability, as consumption and demand emerge from the dynamics of practices (Hesselgren, 2019; Schatzki., 2014; Spurling & McMeekin, 2014). This shift from a user-centred design

approach of identifying and fulfilling perceived needs to instead understanding how needs are created and changed is suggested to be a paradigm shift in design (Kuijer & Bakker, 2015). The concepts prototyped in the DLLs have not mainly been designed to fit current user needs or preferences but to explore how these perceptions can be changed. Importantly, we do not view the users as entirely rational actors or passive consumers but rather as performers of social practices. Instead of informing or guiding individual choice, we have sought to enable or provoke the emergence of new everyday practices. Our work as design researchers has been to support and request experimentation and reflection from our co-researchers. We find that we are often catalysing more than designing change. Viewing practices as dynamic social constructions and needs as malleable offers a better starting point for designing for sustainability transitions (Shove et al., 2007). It is a turn towards prioritising the long-term and 'diffuse' need for a liveable planet over the perceived immediate 'need' for daily conveniences, market-driven and hedonistic impulses.

Co-designing and co-researchers

User involvement in co-design is generally regarded as taking an active part in the design process and influencing decision-making (see, e.g. Kohtala et al., 2020; Kujala, 2003; Steen & van Beuren, 2017). Instead, we have used the term co-researcher to refer to research participants who explore rather than co-design or test (Sjöman, 2023), who are engaged as experts in their lives and experiences (Sanders & Stappers, 2012; Sleeswijk Visser et al., 2005), and in their capacity for reflexivity (Finlay, 2002). In all three cases, the design interventions enabled or provoked the co-researchers to experiment and encouraged them to reflect on their actions, lifestyles, and learnings.

Many transition design approaches build upon engaging the people affected by change in different forms of co-creation. Design visioning and future visualisations are frequently and effectively used to provoke emotional responses and reflections. However, these approaches are often limited to workshop processes, and complementary methods are needed to explore everyday life's social and systemic complexities. In our studies, we found that when people were confronted with future prototypes in their everyday lives, their actions and reflections were more grounded and differed substantially from reactions to hypothetical scenarios (see, e.g. Ilstedt & Sjöman, 2022; Vaddadi et al., 2022).

In design for social innovation, design proposals are often limited to very local user communities (Chen et al., 2016), their perceived needs and current circumstances. Similarly, extended co-design approaches (Hyysalo & Botero, 2013) have successfully included user communities in co-designing services that were better adapted to their *current* everyday practices and social interactions. We find that our design-led and interventionist approach and the explorative co-researcher role have allowed for studying more radical concepts. Overall, we have designed *for* and explored *with* the participants, as they were actively engaged as co-researchers during the experiments rather than co-designers of the interventions.

However, to intervene in people's everyday lives involves ethical considerations. Even if people volunteer to participate, it is hard to foresee what the labs may entail in the form of cumbersome relearning, breakdowns or conflicts with colleagues and housemates. Supporting the participants while allowing for 'breakdowns' of interest from a research point of view is an act of balance of ethics that is always ongoing and significant for this type of research. Still, many of the co-researchers have described the learning experiences as rewarding, which is in line with Scott et al.'s (2012) findings from similar modes of experimentation.

Future prototyping for system-level learnings

In the DLLs the researchers have deliberately designed the future prototypes as provisional tools for learning. Puerari et al. (2018) distinguish between co-creation with the purpose of *making vs learning*, where making aims at a specific innovation goal or output, while learning aims at creating knowledge and enabling changes at a socio-technical system level. For us, this distinction is important because when the aim is learning, experimentation is not limited to product or service development but is open to more general and system-level findings. Many living labs, pilots, and testbeds are restricted by narrow aims and are limited to the current situations. Kommonen and Botero (2013) find that living labs are often producer-driven and geared towards narrow product development processes involving users mainly for testing specific features.

We suggest that open-ended exploration allows the identification of important socio-technical system factors outside of the immediate user relation to a product or service. Experimentation in everyday life has helped identify how societal norms, regulations and technical infrastructures may be unfit for or even bar behaviour change. These findings have provided pathways to promote change on a system level.

References

Auger, J. (2013). Speculative design: crafting the speculation. *Digital Creativity, 24(1),* 11–35.

Botero A. & Hyysalo S. (2013). Ageing together: Steps towards evolutionary co-design in everyday practices, CoDesign, 9:1, 37–54.

Brons, A., van Der Gaast, K., Awuh, H., Jansma, J. E., Segreto, C., & Wertheim-Heck, S. (2022). A tale of two labs: Rethinking urban living labs for advancing citizen engagement in food system transformations. *Cities, 123,* 103552. https://doi.org/10.1016/j.cities.2021.103552

Burns, C., Cottam, H., Vanstone, C., & Winhall, J. (2006). *RED paper 02: Transformation design.* London: Design Council.

Dunne, A. & Raby, F. (2011). *Design Noir: the secret life of electronic objects.* Springer Science & Business Media.

Caniglia, G., Lüderitz, C., von Wirth, T., Fazey, I., Martín-López, B., Hondrila, K., ... & Lang, D. J. (2021). A pluralistic and integrated approach to action-oriented knowledge for sustainability. *Nature Sustainability, 4(2),* 93–100.

Ceschin, F. (2014). *How the design of socio-technical experiments can enable radical changes for sustainability.* Doctoral Dissertation. Brunel University.

Ceschin, F., & Gaziulusoy, I. (2016). Evolution of design for sustainability: From product design to design for system innovations and transitions. *Design studies, 47,* pp. 118–163.

Ceschin, F., & Gaziulusoy, I. (2019). *Design for sustainability: A multi-level framework from products to socio-technical systems.* Routledge.

Chen, D. S., Cheng, L. L., Hummels, C. C. M., & Koskinen, I. (2016). Social design: an introduction. *International Journal of Design*, 10(1), 1–5.

Corsín Jiménez, A. (2014). Introduction. The prototype: More than many and less than one. *Journal of Cultural Economy, 7(4),* pp. 381–398.

Dourish, P., & Button, G. (1998). On 'Technomethodology': Foundational Relationships between Ethnomethodology and System Design. *Human–Computer Interaction 13*: pp. 395–432.

Edwards, G. A., & Bulkeley, H. (2018). Heterotopia and the urban politics of climate change experimentation. *Environment and Planning D: Society and Space, 36(2),* 350–369.

Elzen, B., Geels, F. W., & Green, K. (Eds.). (2004). *System innovation and the transition to sustainability: theory, evidence and policy.* Edward Elgar Publishing.

European Network of Living Labs. (n.d.) What are Living Labs. https://enoll.org/about-us/

Fassi, Davide & Meroni, Anna & Simeone, Giulia. (2013). Design for Social Innovation as a form of Design Activism: An action format. *Conference Proceedings. Social Frontiers 2013, London.*

Finlay, L. (2002). Negotiating the swamp: the opportunity and challenge of reflexivity in research practice. *Qualitative research, 2(2),* 209–230.

Gaziulusoy, A. I., & Ryan, C. (2017). Shifting conversations for sustainability transitions using participatory design visioning. *The Design Journal*, 20(sup1), S1916-S1926.

Geels, F., Turnheim, B., Asquith, M., Kern, F., & Kivimaa, P. (2019). Sustainability transitions: policy and practice. *EEA Report No 9/2019.* European Environment Agency.

Green, K., & Vergragt, P. (2002). Towards sustainable households: a methodology for developing sustainable technological and social innovations. *Futures, 34(5),* 381-400.

Hasselqvist, H., & Hesselgren, M. (2019). Bridging citizen and stakeholder perspectives of sustainable mobility through practice-oriented design. *Sustainability: Science, Practice, and Policy, 15(1), 1–14. https://doi.org/10.1080/15487733.2018.1533781*

Hillgren P-A., Seravalli A., & Emilson A. (2011). Prototyping and infrastructuring in design for social innovation, *CoDesign, 7:3-4,* 169–183. http://dx.doi.org/10.1080/15710882.2011.630474

Hillgren, P.-A. (2013). Participatory design for social and public innovation: Living Labs as spaces for agonistic experiments and friendly hacking. *Public and collaborative: Exploring the intersection of design, social innovation and public policy*, 75–88.

Hesselgren, M. (2023). Designing interventions as learning spaces. In Nordes 2023*: This Space Intentionally Left Blank. 12-14 June, Linköping University, Norrköping, Sweden. https://doi.org.10.21606/nordes.2023.95*

Hesselgren, M. (2019). Designing for sustainability practices: RE-DO design doings, strategies and postures. (Doctoral dissertation, KTH Royal Institute of Technology).

Hesselgren, M., Hasselqvist, H., & Sopjani, L. (2017). Design strategies for exploring and bridging: Intersections of everyday life and decision making for sustainability. *Design Management Academy Conference 2017, Hong Kong 7-9 June 2017. Vol. 1. 2017.*

Hesselgren, M., & Hasselqvist, H. (2016). Give car-free life a try: designing seeds for changed practices. *In DRS2016: Future-Focused Thinking, 27-30 June 2016, Brighton, UK.*

Hossain, M., Leminen, S., & Westerlund, M. (2019). A systematic review of living lab literature. *Journal of cleaner production, 213,* pp. 976–988.

Ilstedt, S., & Wangel, J. (2014). Altering expectations: How design fictions and backcasting can leverage sustainable lifestyles. In *DRS (Design Research Society) 2014: Design's Big Debates-Pushing the Boundaries of Design Research.* Umeå, Sweden, 2014.

Ilstedt, S., and Sjöman, M. (2022). The value of being close: Social and ecological sustainability in co-living for students, in Lockton, D., Lenzi, S., Hekkert, P., Oak, A., Sádaba, J., Lloyd, P. (eds.), D*RS 2022: Bilbao*, 25 June - 3 July, Bilbao, Spain. https://doi.org/10.21606/drs.2022.413

Irwin T., Kossoff G. & Tonkinwise C. (2015). Transition Design Provocation. *Design Philosophy Papers, 13(1),* 3–11. http://dx.doi.org/10.1080/14487136.2015.1085688

Kimbell, L., & Bailey, J. (2017). Prototyping and the new spirit of policy-making. *CoDesign, 13(3),* 214–226.

Kommonen, K. H., & Botero, A. (2013). Are the users driving, and how open is open? Experiences from Living Lab and user driven innovation projects. *The Journal of Community Informatics, 9(3).*

Kohtala, C., Hyysalo, S., & Whalen, J. (2020). A taxonomy of users' active design engagement in the 21st century. *Design Studies, 67,* 27–54.

Kuijer, L. (2014). Implications of Social Practice Theory for Sustainable Design. (Doctoral dissertation)

Kuijer, L., & Bakker, C. (2015). Of chalk and cheese: behaviour change and practice theory in sustainable design. *International Journal of Sustainable Engineering, 8(3), 219–230.*

Kujala, S. (2003). User involvement: a review of the benefits and challenges. *Behaviour & information technology*, 22(1), 1–16.

Leminen, S., Westerlund, M., & Nyström, A. G. (2014). On becoming creative consumers–user roles in living labs networks. *International Journal of Technology Marketing, 9(1),* 33–52.

Liedtke, C., Baedeker, C., Hasselkuß, M., Rohn, H., & Grinewitschus, V. (2015). User-integrated innovation in Sustainable LivingLabs: An experimental infrastructure for researching and developing sustainable product service systems. *Journal of Cleaner Production, 97,* 106–116. https://doi.org/10.1016/j.jclepro.2014.04.070

Lilley, D. (2009). Design for sustainable behaviour: strategies and perceptions. *Design Studies, 30,* pp. 704–720.

Lockton, D., Harrison, D., & Stanton, N. (2008). Making the user more efficient: design for sustainable behavior. *International Journal of Sustainable Engineering*, 1, pp. 3–8.

Loorbach, D. (2007). *Transition management. New mode of governance for sustainable development.* Utrecht: International Books.

Lähteenoja, S., Marttila, T., Gaziulusoy, İ., & Hyysalo, S. (2023). Transition co-design dynamics in high level policy processes. *Design Studies, 88,* 101207.

Malmqvist, T. and Brismark, J. (2023). Embodied carbon savings of co-living and implications for metrics. *Building and Cities, 4 (1),* pp 386–404.

Manzini, E. (2015). *Design, when everybody designs: An introduction to design for social innovation.* MIT Press.

Manzini, E., & Rizzo, F. (2011). Small projects/large changes: Participatory design as an open participated process. *CoDesign, 7(3-4),* pp. 199-215.

Marres, N., & Stark, D. (2020). Put to the test: For a new sociology of testing. *The British Journal of Sociology, 71(3),* 423–443.

Marvin, S., Bulkeley, H., Mai, L., McCormick, K., & Palgan, Y. V. (2018). Urban living labs: Experimenting with city futures. In *Urban Living Labs: Experimenting with City Futures*. https://doi.org/10.4324/9781315230641

McCrory, G., Schäpke, N., Holmén, J., & Holmberg, J. (2020). Sustainability-oriented labs in real-world contexts: An exploratory review. *Journal of Cleaner Production, 277, 123202.*

Puerari, E., De Koning, J. I., Von Wirth, T., Karré, P. M., Mulder, I. J., & Loorbach, D. A. (2018). Co-creation dynamics in urban living labs. *Sustainability, 10(6), 1893.*

Reckwitz, A. (2002). Toward a theory of social practices: a development in culturalist theorizing. *Eur. J. of Social Theory 5(2).* https://doi.org/10.1177/13684310222225432.

Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Schatzki, T. (2014). Practices, governance and sustainability. In *Social practices, intervention and sustainability* (pp. 15–30). Routledge.

Scott, K., Bakker, C., & Quist, J. (2012). Designing change by living change. *Design Studies*, *33*(3), 279–297. https://doi.org/10.1016/j.destud.2011.08.002

Sengers, F. Wieczorek, A J. Raven, R. (2019) Experimenting for sustainability transitions: A systematic literature review. *Technological Forecasting and Social Change, Volume 145, 2019,* Pages 153–164.

Shove, E. (2003). Converging conventions of comfort, cleanliness and convenience. *Journal of Consumer Policy, 26,* pp. 395–418.

Shove, E., & Watson, M. (2006). *POPD Manifesto*. Durham University.

Shove, E., Watson, M., Ingram, J., & Hand, M. (2007). *The design of everyday life.* Oxford: Berg. https://doi.org/10.5040/9781474293679

Sjöman, M., and Hesselgren, M. (2020). Designerly Living Labs: Early-stage exploration of future sustainable concepts, in *Boess, S., Cheung, M. and Cain, R. (eds.), Synergy - DRS International Conference 2020, 11-14 August*, Held online. https://doi.org/10.21606/drs.2020.307

Sjöman, M. (2023). *Living the Change: Designerly modes of real-life experimentation.* Doctoral dissertation, KTH Royal Institute of Technology.

Sleeswijk Visser, F. S., Stappers, P. J., Van der Lugt, R., & Sanders, E. B. (2005). Contextmapping: experiences from practice. *CoDesign, 1(2),* 119–149.

Spaargaren, G. (2011). Theories of practices: Agency, technology, and culture: Exploring the relevance of practice theories for the governance of sustainable consumption practices in the new world-order. *Global Environmental Change, 21(3),* pp. 813–822. https://doi.org/10.10167j.gloenvcha.2011.03.010

Spurling, N., & McMeekin, A. (2014). Interventions in practices: Sustainable mobility policies in England. *In Social practices, intervention and sustainability* (pp. 78–94). Routledge.

Steen, K., & Van Bueren, E. (2017). The defining characteristics of urban living labs. *Technology Innovation Management Review, 7(7).*

Sterling, B. (2009). Design Fiction, *Interactions 16,* p 16–20, ACM press.

Tukker, A., Haag, E., Eder, P., Vercalsteren, A., Tischner, T. W. U., Charter, M., ... & van der Vlugt, M. (2000). Eco-design: European state of the art. *Seville, Spain: Institute for Prospective Technological Studies.*

Vaddadi, B., Ringenson, T., Sjöman, M., Hesselgren, M., & Kramers, A. (2022). Do they work? Exploring possible potentials of neighbourhood Telecommuting centres in supporting sustainable travel. *Travel Behaviour and Society, 29,* 34–41.

Voytenko, Y., McCormick, K., Evans, J., & Schliwa, G. (2016). Urban living labs for sustainability and low carbon cities in Europe: Towards a research agenda. *Journal of Cleaner Production, 123,* pp. 45–54

About the Authors:

Mia Hesselgren is an assistant professor in industrial design engineering at KTH Royal Institute of Technology, Stockholm. As a transdisciplinary design researcher, she uses practice-oriented approaches, and her research involves interventions in everyday life through designed systems that reconfigure practices.

Sara Ilstedt is a professor in product and service design at KTH Royal Institute of Technology. She has worked with wellbeing and sustainable lifestyle changes since 2004. 2012, she started the Green-Leap research network using design to catalyse sustainability change.

Martin Sjöman is a researcher and design lecturer at KTH Royal Institute of Technology. Martin holds an MFA in Industrial Design, and his research focuses on design practice as a driver of system transformation and interdisciplinary collaboration.